

Metric Multidimensional Scaling

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Introduction

Metric multidimensional scaling — also known as classical MDS or principal coordinates analysis (PCoA) — takes a pairwise distance matrix and produces a low-dimensional Euclidean configuration whose pairwise distances approximate the input as closely as possible. It is the workhorse for visualising similarity or distance data when raw coordinates are unavailable: think reconstructing a map from inter-city distances, projecting microbial-community Bray-Curtis dissimilarities, or visualising a kernel-derived similarity matrix in a 2D plot.

Prerequisites

A working understanding of distance and similarity matrices, principal components analysis, and basic linear algebra (eigendecomposition).

Theory

Given an $n \times n$ distance matrix D , form the doubly-centred inner-product matrix

$$B = -\frac{1}{2}(I - J/n)D^{(2)}(I - J/n),$$

where $D^{(2)}$ has elements d_{ij}^2 and J is the all-ones matrix. Eigendecompose $B = U\Lambda U^\top$. The top k eigenvectors scaled by $\sqrt{\lambda}$ give a k -dimensional configuration whose Euclidean inter-point distances approximate the original distances. When D comes from Euclidean distances on raw data, classical MDS is equivalent to PCA on the centred data — the eigenvalues match the squared singular values of the centred data matrix.

For non-Euclidean distances, B may have negative eigenvalues; the configuration still uses only the top positive eigenvalues, with the discarded negative mass quantifying the embedding error.

Assumptions

A pairwise distance matrix that is at least approximately Euclidean-embeddable. Severe violations (negative eigenvalues dominating) suggest non-metric MDS as a more appropriate alternative.

R Implementation

```
# Classical MDS on US city distances
data(UScitiesD)
mds <- cmdscale(UScitiesD, k = 2, eig = TRUE)
plot(mds$points, type = "n")
text(mds$points, labels = labels(UScitiesD))

# Eigenvalues and variance explained
mds$eig[1:5]
cumsum(mds$eig[1:5] / sum(mds$eig[1:5]))
```

Output & Results

`cmdscale()` returns the low-dimensional configuration (`$points`), eigenvalues, and an indicator of how well the chosen k approximates the original distances. The plot of US-city MDS coordinates recovers the recognisable geographical structure with a near-perfect rotation and reflection of the actual map.

Interpretation

A reporting sentence: “Classical MDS on the 10 inter-city distances produced a two-dimensional configuration that captured 99 % of the eigen-mass; the resulting layout matches actual geography after a reflection, with east-west and north-south structure immediately visible.” Always report the proportion of eigen-mass retained in the displayed dimensions; a poor approximation signals that more dimensions or non-metric MDS would help.

Practical Tips

- For Euclidean distances, classical MDS is equivalent to PCA on the centred data; if you have raw data, run PCA directly for cleaner output.
- For non-Euclidean distances (Bray-Curtis, Jaccard, network distances), check the eigenvalue structure — large negative eigenvalues indicate a non-Euclidean metric and motivate non-metric MDS instead.
- In ecology and microbiome research, this method is called principal coordinates analysis (PCoA); the algorithm is identical.
- Add Procrustes analysis (`vegan::procrustes`) when comparing MDS solutions across related datasets; it aligns configurations up to rotation, reflection, and scaling.
- For configurations beyond two dimensions, plot pairs of axes; `factoextra::fviz_mds()` provides a ggplot-style alternative.
- Pair the configuration with a stress measure or the proportion of variance retained to communicate fit quality alongside the visual.

R Packages Used

Base R `cmdscale()` for classical MDS; `vegan::wcmdscale()` for weighted classical MDS in ecology; `MASS::isoMDS()` and `vegan::metaMDS()` for non-metric alternatives; `factoextra::fviz_mds()` for ggplot-style visualisation; `smacof` for advanced MDS variants including weighted and constrained versions.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE